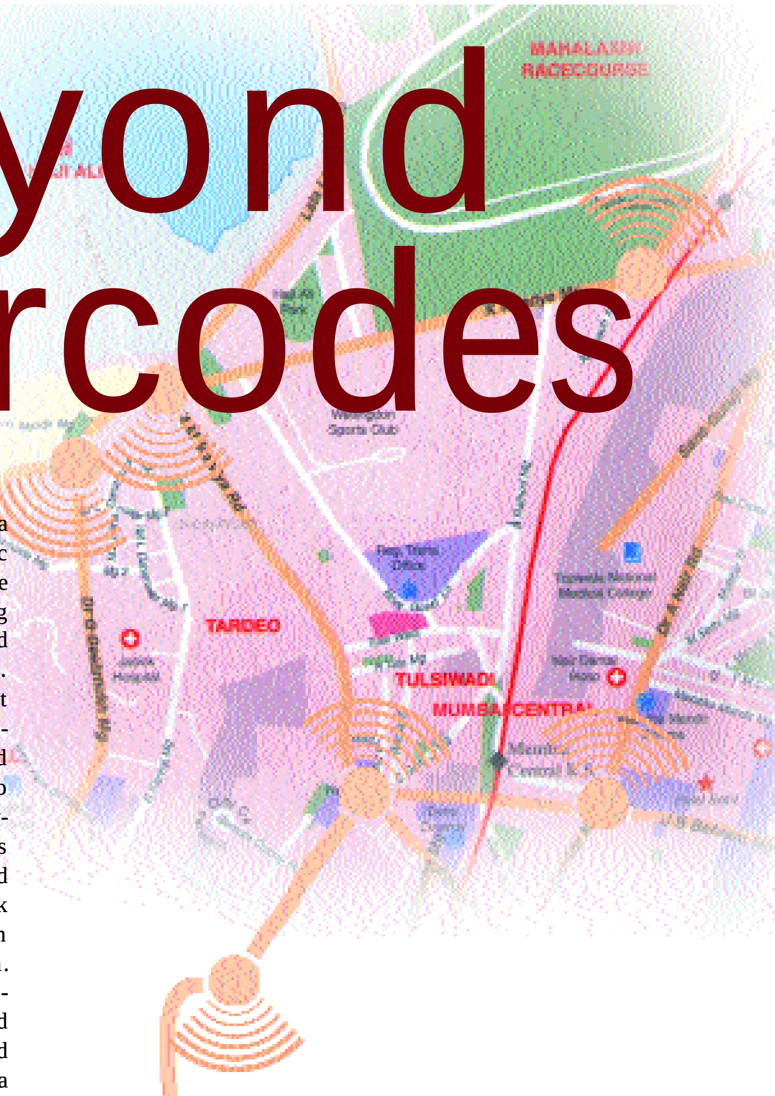
Beyond Barcodes



Someone's taking those little black and white bars on your bag of chips further...

You just can't beat a great idea. The basic barcode, or the Universal Packaging Code (UPC), first introduced in the early 1970s, is one such. UPC barcoding was the first standardized product identification system. It's simple and easy to implement: with no effort, you can barcode everything from groceries to books and clothes. The white and black bars require just a quick scan to update the status on the inventory tracking system.

The idea is to allow manufacturers, wholesalers and retailers track products and shipments. Whether it's a supermarket checkout line or your local British Council library, barcodes have found use in all kinds of places.

Intelligent stripes

Radio Frequency Identification (RFID), married with today's Enterprise Resource Planning (ERP) solutions, may soon be the next big thing in inventory management. The technology provides an automated method to collect product or transaction information. The RFID system works using 'smart' tags, with inbuilt silicon chips that store data, a reader that scans information from the tags, and the infrastructure to store and analyse the data.

Barcodes made inventory tracking easier, but they have their disadvantages. For starters, each barcode has to be read in individually by the reader. If an entire warehouse-worth of products needed to be checked into a computer, it could mean several hours of work.

Furthermore, the basic barcode is just a tag with data printed on it, and this data can't be updated. Except, of course, by sticking another barcode over it.

This is where RFID tags come in. RFID takes asset tracking to the next level; with smart, intelligent tags embedded in the package, the information on the tag can be scanned and updated automatically by 'readers'. Little wonder that courier service FedEx, and US retail giant Wal-Mart, are looking at RFID implementation for streamlining delivery and asset management.

How RFID works

An RFID smart tag has a tiny inbuilt integrated circuit, and a coiled antenna. A tag beams out the information it carries—how far the tag can beam information depends on the operating frequency—and automated reading can be done by installing readers wherever required.

The readers are connected to a computer network. After a reader reads the data from the tag, it passes it to the computing network associated with the RFID system. The network takes this incoming data, and tracks and updates its records. If the tag information needs to be updated, appropriate instructions are sent back to the tags.

Tags are of two types—active and passive. An active tag has its own, internal power supply. This allows them, for instance, to add in new data or modify existing data, independent of a reader. Passive tags have no internal power supply; the same electromagnetic field that is emitted by the reader is used to power up the tag, so that it talks back to the reader.

The advantage is that passive tags can have nearly unlimited life spans, and need not be out of service unless there's a circuitry breakdown. Active tags, though, have a lifespan limited by the

Solomon Lewis